

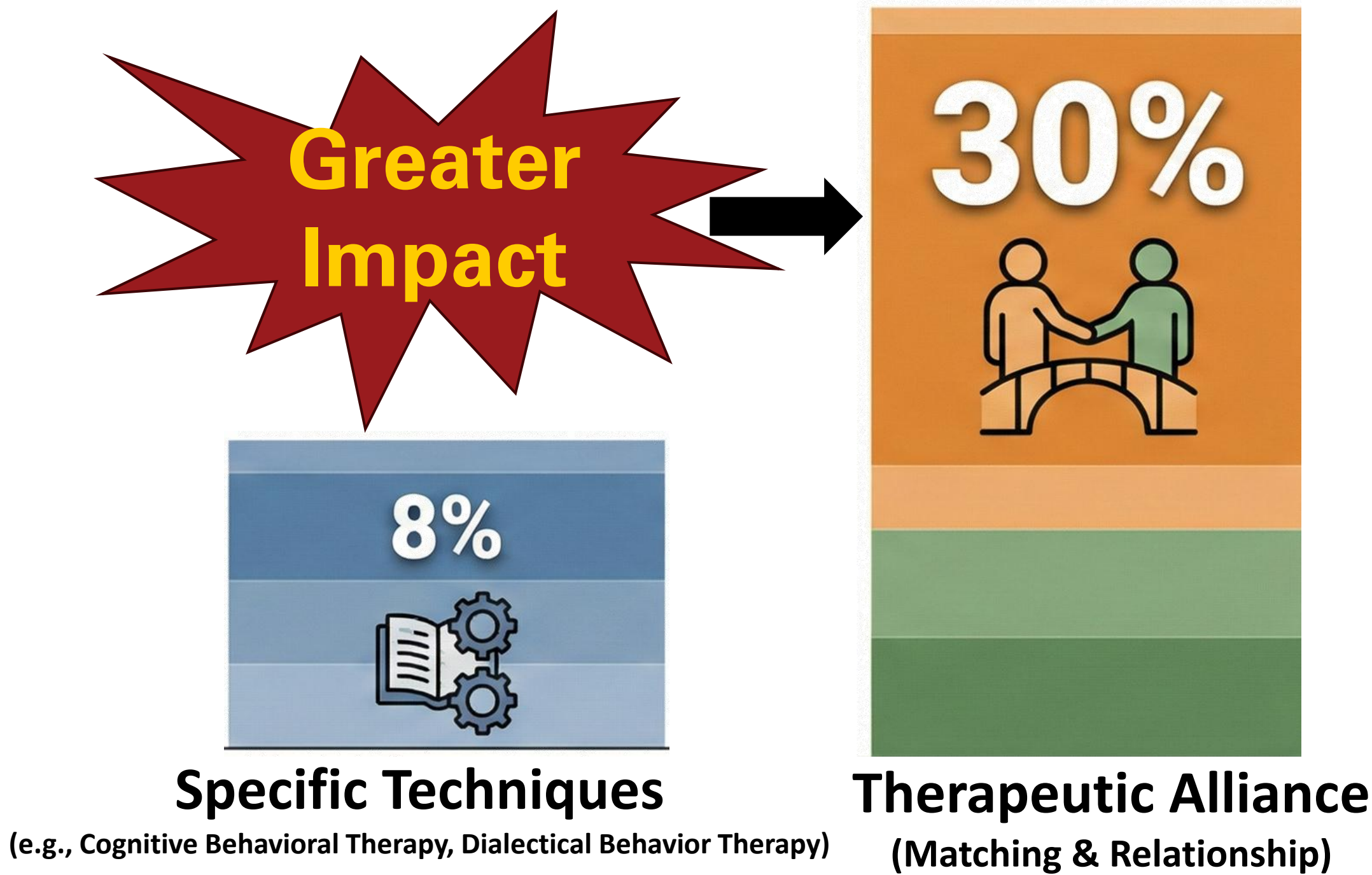
High Cost of Mismatch

Current intake models are inefficient, leading to massive resource waste and poor outcomes.

- HIGH DROPOUT RATES** [1-3] :
~20% to 35% of clients terminate therapy prematurely in real-world settings.
- THE CRITICAL "FIRST JUMP"** [1-3] :
Up to 50% of dropouts occur after just the 1st Session due to poor initial fit.
- ECONOMIC BURDEN** [4]:
\$16 Trillion projected global economic loss by 2030 due to mental disorders.

Why Matching Matters?

Evidence proves that WHO the therapist is matters more than what technique they use [5].



VELUTI: Multimodal Client-Therapist Matching Framework

Phase I: Initial Surveys

- Client Side: Standardized Baseline Assessments**
- Completion of validated self-report measures (e.g., PHQ-9, World Health Organization (WHO) forms) to establish initial symptom severity and subjective well-being profiles
- Therapist Side: Professional Profiling & Simulation Configuration**
- Surveys detailing clinical expertise and specialties, alongside topic-specific questionnaires used to configure subsequent simulated agent interactions

Phase 2: Multimodal Data Collection via Interaction with Virtual Agents

- Engages clients in empathetic intake dialogues co-designed with licensed professionals
- Systematically captures essential information, including presenting topics and primary motivations for seeking help

Client Interaction



(Virtual Therapist Agent)

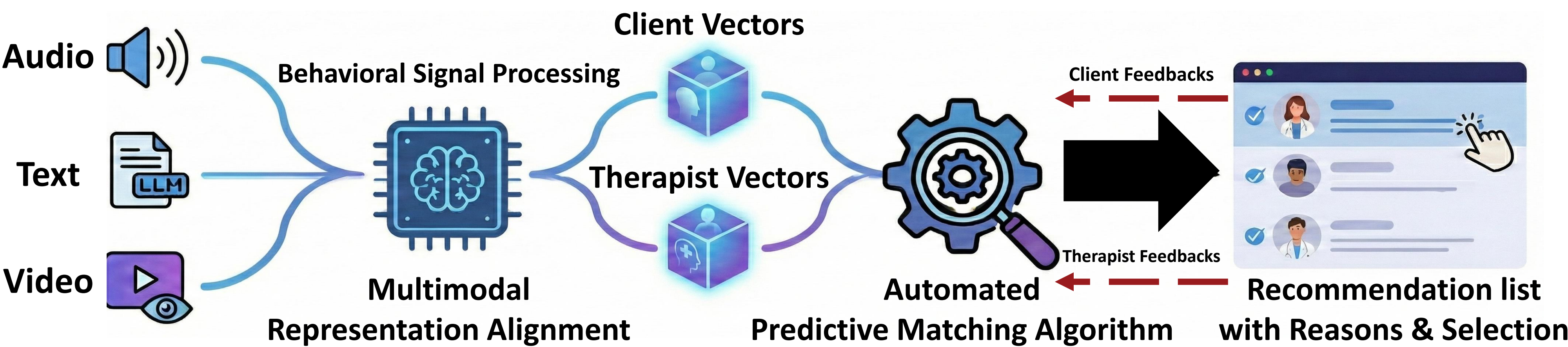
Therapist Interaction



(Virtual Client Agent)

- Provides realistic, simulated clinical scenarios for standardized behavioral assessment
- Captures multifaceted professional profiles, including therapeutic style, communication approach; paralinguistic voice features

Phase 3: Client-In-The-Loop Predictive Matching Models



[1] Swift, J. K., & Greenberg, R. P. (2014). A treatment by disorder meta-analysis of dropout from psychotherapy. *Journal of Psychotherapy Integration*.
[2] Swift, J. K., et al. (2017). Treatment refusal and premature termination in psychotherapy, pharmacotherapy, and their combination: A meta-analysis of head-to-head comparisons. *Psychotherapy*.
[3] Simon, G. E., & Ludman, E. J. (2010). Predictors of early dropout from psychotherapy for depression in community practice. *Psychiatric Services*, 61(7), 684-689.
[4] Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., ... & Unützer, J. (2018). The Lancet Commission on global mental health and sustainable development.
[5] Wampold, B. E., & Imel, Z. E. (2015). *The great psychotherapy debate: The evidence for what makes psychotherapy work*. Routledge.

